S4 Text.

Earlier EvoSim results and the audit trail

Supporting information for "Environmental fluctuation limits gene regulation unless it can be tracked"

This supplement records the EvoSim results that preceded the present study, what became of each, and the measurement errors found along the way together with the check that exposed each one. The complete record, including every retracted analysis, is in the project's `docs/EvoSim.md` and `PAPER.md`.

1. Earlier EvoSim results and their status

These results came from experiments with smaller populations (mostly between 2,000 and 3,000 organisms) than the 180-run batch in S6 Text.

Earlier result	Evidence	Status after this study
Fluctuation increases the number of network nodes	7 of 7 frozen–fluctuating comparisons positive and significant; mean +6.32 nodes (range +2.64 to +10.15)	Reproduces published results [1,2]. R-Aevol shows the opposite (P4), at every grain tested (G1–G4), so this is a difference between the models.
Fluctuation increases connection density	Count density +0.09 to +0.14, significant in 5 of 6 comparisons; the edge increase exceeded that expected from node number alone by 27 to 41 percentage points; stable across development iterations 4 to 20	The count effect replicates (+0.12 at grid 128), but strength density falls by 11% and total strength does not change. The interpretation as denser wiring is withdrawn.
The information content of the cue does not matter	Pooled over 15 informative–scrambled comparisons, mean difference −0.17 nodes, $t = -0.30$, sign test $p = 0.61$	For genome size, reproduced in R-Aevol, where informative and yoked cues gave similar numbers of functional genes. The informative cue did change regulation in R-Aevol (P2), a measure not taken in the EvoSim regime in which organisms read the cue.
Wiring cost makes fluctuation lethal	In the sparsest regime tested, 0 of 60 frozen and 37 and 33 of 60 fluctuating populations went extinct (two-sided Fisher exact $p = 4 \times 10^{-15}$ and 5×10^{-13})	Consistent with the viability pilot for P5: fluctuating populations survived 2, 2 and 5 of 6 times at grids 80, 96 and 112, against 6 of 6 frozen.
Task difficulty does not set genome size	Four manipulations of the task, all null (detection floors 2.2 to 4.2 nodes)	Stands.
Organisms read the cue when development is reversible	With living organisms re-developed every few steps, switchers 0.645 against 0.229, and the best-paying channel matched in 1.5 against 1.0 of 6 states (informative against scrambled, 100 populations per arm); genome size unchanged ($p = 0.84$, 95% CI −2.1 to +2.6 nodes)	Stands. The strength measure has not been taken in this regime.

Earlier result	Evidence	Status after this study
Long-run outcome	8 runs of 25 million steps (about 630,000 generations): node number fell to the minimum (12 at the time) or within two nodes of it by generation 14,000 and stayed there, and six of the eight runs ended exactly at the minimum	Stands; the present version has a minimum of 13 because a sensor node was added.

2. Record of pre-registered predictions

Throughout the project, predictions were committed to version control before the runs that tested them.

Prediction	Outcome
Earlier EvoSim work	
Genome size rises with the complexity of the task	wrong (no change)
Rotating the yields alone makes the task conditional	wrong (inflow dominated)
An abiotic cue will be read because it cannot be depleted	wrong (legible but unread)
A conditional genome is adaptive but hard to reach by mutation	wrong (selected against)
Reversible development makes a fine-grained regime habitable	correct (survival rose from 0 of 3 to 3 of 3)
A conditional response, once evolved, raises genome size	wrong (the response evolved, size did not change)
This study	
P1–P4 (R-Aevol sweep)	all supported
P5 (EvoSim confirmation)	not run (no viable regime)
G1, G3 (grain, coarse against fine)	inconclusive
G2, G4 (grain, coarse against frozen)	refuted
C1–C4 (R-Aevol confirmation of the signal-link decomposition)	all supported
W1, W2 (R-Aevol environment switch)	existing networks lost; networks grow once fluctuation stops

3. Measurement errors and the checks that found them

Almost every wrong conclusion in the project traced back to an error of measurement, and each was found by a check whose correct answer was known in advance. They are listed because the same errors recur in simulation studies.

Error	What it produced	How it was found
Edge counts used as a measure of wiring	A density increase that is not an increase in wiring strength (this study)	Measuring summed absolute weight, prompted by the R-Aevol replication
A statistic read without checking its definition	`mean_abs_w` is summed weight divided by all n_g^2 slots, so multiplying it by the edge count gave meaningless "total strength" columns	Reading the source of the statistic before deriving from it
Parameters attributed to the wrong source	A tenfold error in the stated mutation rate of R-Aevol's configuration, in a draft of the Methods	Comparing against the platform's example files

Error	What it produced	How it was found
A control that was not a control	Turning off the yield rotation also made the world poorer, and the arm with the least conditional task had the largest genomes	Measuring population size and energy in every arm
A no-op manipulation that was not a no-op	Removing a node's inputs changed its output downstream and altered every trait (repair 0.995 to 0.127)	Requiring a zero-effect arm whose result is known
Tags that leaked	Tagging organisms by storage slot, which is reused after death, spread the tag to unrelated organisms	A control arm that reached fixation when it should not
Wrong unit of analysis	Pooling repeated measurements of the same node inflated a correlation	Reducing each node to one value
Survivorship in the sampling frame	Sampling only at the end of runs, when almost all nodes were old	Sampling during runs
A derived timescale from the wrong statistic	Generations per environmental state computed from a survivor-biased lifespan, a threefold error	Using the slope of lineage depth
Small samples	Three promising signals from 3 or 4 seeds, all gone at 12 seeds	Replication at larger n
A control checked in one mode only	The scrambled cue leaked 0.067 bits of information without a spatial gradient, the mode mostly used	Measuring its information content in every mode; fixed to 0.0004 bits
A regime that kills one arm	At grid 48, 6 of 6 fluctuating and 0 of 6 frozen populations died, which would have made the comparison one of survivors	A viability check at full run length before any batch
Identical arms counted as replicates	With a single state the cue is zero, so the three frozen arms were the same runs	Checking that arms differ
A verification script that could not fail	A scoping error kept an accumulator at its initial value, so the script would have reported success whatever the data	A negative control with a known failure
A significance test against the wrong spread	Dividing a difference by one arm's standard deviation instead of the standard error of the difference reported "no effect" for $t = 4.2$	Recomputing with Welch and permutation tests

References

1. Tsuda ME, Kawata M. Evolution of gene regulatory networks by fluctuating selection and intrinsic constraints. PLoS Comput Biol. 2010;6(8):e1000873.

2. Bentkowski P, van Oosterhout C, Mock T. A model of genome size evolution for prokaryotes in stable and fluctuating environments. Genome Biol Evol. 2015;7(8):2344–51.